Cell Site & Transceiver Data Pre-processing & Upload Guidelines (SOP)

This document provides a comprehensive operational guide for data uploaders to sanitize, format, and validate Cell Site and Base Transceiver Station (BTS) Excel files before uploading them to the QoS portal. Adhering to these specifications guarantees proper spatial mapping, RF propagation modeling, and prevents processing failures.

> **Multi-Row Header Structure:** The Cell Site template utilizes a specialized **3-row composite header** (Row 1: Generation, Row 2: Technology, Row 3: Band / Property). **Data rows begin strictly on Row 4**. Never delete, merge, reorder, or alter the header rows.

1. Summary Checklist for Uploaders

- ☐ Template maintains the exact 3-row header hierarchy; data rows begin on **Row 4**.
- ☐ Required location and provider fields (`Provider` , `Municipality` , `Province` , `Region` , `Lat` , `Long`) are fully populated.
- ☐ **Active Technology Rule:** At least ONE band column is set to `1` or `TRUE` (or contains valid center frequency `CF: <val>, BW: <val>`). Rows where all band columns are `0` , `FALSE` , or empty/blank cannot be used for coverage analysis.
- ☐ `Provider` matches one of the registered **Wireless Service Providers** (e.g., `DITO TELECOMMUNITY CORP.` , `GLOBE COMMUNICATIONS INC.` , `SMART COMMUNICATIONS INC.`).
- ☐ Coordinates (`Lat` , `Long`) contain clean decimal numbers in WGS84 format without degree symbols or text letters.
- ☐ `Tower Type` and `AntennaType` use exact approved dropdown values.
- ☐ `Antenna Height (m)` is a pure positive numeric value (meters, no `"m"` unit).
- ☐ `Azimuth (°)` is a valid heading between `0` and `360` degrees (or semicolon-delimited for multi-sectors, e.g., `0; 120; 240`).
- ☐ `Beam Tilt (°)` contains numeric tilt angles only (must match Azimuth sector count if semicolon-delimited, e.g., `2; 2; 2`).
- ☐ Empty optional cells are left completely **BLANK** (no `"---"` , `"N/A"` , or `"null"` placeholders).

2. Header Structure & Data Start Row

Row Number	Level	Description
Row 1	**Generation**	Defines cellular generations: `2G` , `3G` , `4G` , `5G` , plus antenna parameters
Row 2	**Technology**	Defines air interface standards: `GSM` , `WCDMA` , `FDD-LTE` , `TDD-LTE` , `NR`
Row 3	**Band / Column**	Core site info, specific band codes (`B8-900` , `B3-1800` , `B28-700` , `B78-3500`), and RF properties

Row Number	Level	Description
Row 4+	**Data Rows**	**First actual site record begins on Row 4**

> **Caution:** If data is mistakenly pasted starting on Row 1, 2, or 3, the entire file parser will fail immediately due to header mismatch.

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3. Required Fields & Mandatory Core Columns

Header Name (Row 3)	Type	Description	Action Required if Missing
`Provider`	Dropdown	Registered wireless telecommunications provider	Select from approved wireless telco dropdown
`Municipality`	Text	City / Municipality of cell tower site	Provide official LGU municipality name
`Province`	Text	Province of cell tower site	Provide official province name
`Region`	Text	Administrative Region (e.g., `Region X`)	Provide official region designation
`Lat`	Decimal	WGS84 Latitude coordinate	Ensure valid decimal format; delete row if unavailable
`Long`	Decimal	WGS84 Longitude coordinate	Ensure valid decimal format; delete row if unavailable
Band Columns	Boolean / Numeric	Operating frequency bands (Columns I to AD)	**At least ONE band must be set to `1` or `TRUE`**

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4. Critical Active Technology Rule (Coverage Analysis)

Every cellular tower sector represents RF transceivers transmitting on specific licensed spectrum bands.

The Rule: At least ONE band column per row must be set to `1` or `TRUE` (or valid CF/BW configuration).

Data State in Row	System Processing Behavior	Coverage Impact
At least one band is `1` or `TRUE` (e.g., `B28-700 = 1`)	• Creates `Cell` (Tower site) • Creates `CellSector` (Antenna structure) • Creates `Cellular` (Active RF transceivers) • Auto-generates `Network` & `Sector`	✅ **Fully Functional:** Tower appears on coverage maps, wireless assessments, and connectivity analytics.

Data State in Row	System Processing Behavior	Coverage Impact
ALL bands are `0`, `FALSE`, or empty/blank	• Creates `Cell` (Tower site) • Creates `CellSector` (Antenna structure) ❌ **Does NOT create Cellular records** ❌ **Does NOT create Network or Sector records**	⚠️ **Degraded State:** Tower site is saved in inventory, but **CANNOT** be analyzed, visualized on heatmap layers, or used for signal coverage calculations.

> **Warning:** If your upload logs display: *"Pre-validation Warning: Found X records with ALL technology/band columns set to FALSE or empty"*, those rows will not produce coverage analysis data. Verify the spreadsheet and set active bands to `1` or `TRUE` before uploading.

4.1 Band Input Options: Presets vs. Custom CF/BW Values

You can indicate active transceiver bands using either **Boolean Presets** or **Custom RF Parameters (CF/BW)**:

Option A: Presets (`1`, `0`, `TRUE`, `FALSE`, or Blank) — Recommended

- **Active Band:** Enter `1` or `TRUE`. The system automatically applies the standardized national regulatory Center Frequency and Channel Bandwidth defaults for that band.
- **Inactive / Unequipped Band:** Enter `0`, `FALSE`, or leave the cell completely **BLANK**.
- *Note:* Matches the exact preset options used in the Wireless Validation module.

Option B: Custom Center Frequency & Bandwidth (`CF: <val>, BW: <val>`)

When an operator utilizes specific carrier frequency assignments or custom channel bandwidths, you can enter the exact values directly in the band cell:

- `CF` **(Center Frequency):** Downlink Center Carrier Frequency in **Megahertz (MHz)**.
- `BW` **(Bandwidth):** Allocated RF Channel Bandwidth in **Megahertz (MHz)**.
- **Syntax Rule:** `CF: <frequency>, BW: <bandwidth>` *(Case-insensitive; spaces after colon and comma are optional).*

Valid Input Examples for Band Columns:

Technology & Band Code	Example Input	Interpreted State	Resulting RF Parameters
4G LTE B28 (700 MHz)	`1` or `TRUE`	Active	Default DL Center Freq: `780.0 MHz`, Bandwidth: `20.0 MHz`
4G LTE B28 (700 MHz)	`CF: 780.5, BW: 20`	Active	DL Center Freq: `780.5 MHz`, Bandwidth: `20.0 MHz`
4G LTE B3 (1800 MHz)	`CF: 1805, BW: 20`	Active	DL Center Freq: `1805.0 MHz`, Bandwidth: `20.0 MHz`
4G LTE B1 (2100 MHz)	`CF: 2140, BW: 15`	Active	DL Center Freq: `2140.0 MHz`, Bandwidth: `15.0 MHz`

Technology & Band Code	Example Input	Interpreted State	Resulting RF Parameters
4G LTE B40 (2300 MHz TDD)	`CF: 2350, BW: 20`	Active	DL Center Freq: `2350.0 MHz`, Bandwidth: `20.0 MHz`
4G LTE B41 (2500 MHz TDD)	`CF: 2595, BW: 20`	Active	DL Center Freq: `2595.0 MHz`, Bandwidth: `20.0 MHz`
5G NR n78 (3500 MHz C-Band)	`CF: 3500, BW: 100`	Active	DL Center Freq: `3500.0 MHz`, Bandwidth: `100.0 MHz`
2G GSM B8 (900 MHz)	`CF: 935.2, BW: 0.2`	Active	DL Center Freq: `935.2 MHz`, Bandwidth: `0.2 MHz` (200 kHz)
Inactive / Not Deployed	`0`, `FALSE`, or *(blank)*	Inactive	Band is skipped; no transceiver record created

5. Dropdown Columns Validation

Column Header	Allowed Values	Remarks
`Provider`	`DITO TELECOMMUNITY CORP.`, `GLOBE COMMUNICATIONS INC.`, `SMART COMMUNICATIONS INC.`	Select strictly from the dropdown list.
`Tower Type`	`GuyedTower`, `Monopole`, `Rooftop`, `SelfSupporting`, `Others`, `Unknown`	Select physical structural tower type.
`AntennaType`	`OmniDirectional`, `Directional`, `Others`, `Unknown`	Select antenna radiation pattern type.
Band Columns (Cols 9 to 30)	`1`, `0`, `TRUE`, `FALSE`, blank, or `CF: <val>, BW: <val>`	• Use `1` or `TRUE` for standard active deployment. • Use `CF: <val>, BW: <val>` for custom carrier frequency. • Use `0`, `FALSE`, or leave blank if not equipped.

6. Numeric & RF Parameters Sanitization

Field	Example Valid Value	Invalid Entries (Will Fail)	Rule
`Lat`	`8.48222`	`8°28'56"N`, `8,48222`, `N 8.48`	Clean decimal latitude only
`Long`	`124.64722`	`124°38'50"E`, `124,64722`	Clean decimal longitude only
`Antenna Height (m)`	`35` or `42.5`	`35m`, `35 meters`, `~35`	Numeric height in meters only

Field	Example Valid Value	Invalid Entries (Will Fail)	Rule
`Azimuth (°)`	`120` (or `0; 120; 240`)	`120 deg`, `120°`, `SE`, `0, 120, 240`	Directional angle (0–360°); use semicolon `;` for multi-sectors
`Beam Tilt (°)`	`2` (or `2; 2; 2`)	`2 degrees`, `2 deg`, `2, 2, 2`	Mechanical/electrical tilt angle; count must match Azimuth

7. Multiple Azimuth and Beam Tilt Formatting (Multi-Sector / Tri-Sector Sites)

In cellular network deployments, a single physical cell tower structure typically hosts multiple directional antenna sectors (e.g., standard 3-sector tri-pole antennas oriented at 120° intervals, or dual-sector configurations).

The QoS portal ingestion engine supports defining multiple sectors directly within a single row by using a **semicolon (`;`) delimiter**.

Strict Semicolon Formatting Rules:

1. **Delimiter Must Be Semicolon (`;`):** Use semicolons `;` (with or without spaces) to separate sector values (e.g., `0; 120; 240` or `0;120;240`). ❌ Do NOT use commas, slashes, or colons.
2. **1-to-1 Count Matching is Strictly Required:** The number of semicolon-separated values in `Azimuth (°)` **must exactly equal** the number of values in `Beam Tilt (°)`. If count mismatches (e.g. 3 azimuths vs 2 beam tilts), sector creation is skipped.
3. **Pure Numeric Values Only:** Each entry must parse cleanly to a decimal number. Do NOT include degree symbols (`°`) or units.

Sector Format Examples & System Behavior:

Antenna Configuration	`Azimuth (°)`	`Beam Tilt (°)`	Sector Breakdown & Ingestion Behavior
Standard Tri-Sector (3 Sectors)	`0; 120; 240`	`2; 2; 2`	✅ **Creates 3 CellSectors:** • Sector 1: Azimuth 0°, Tilt 2° • Sector 2: Azimuth 120°, Tilt 2° • Sector 3: Azimuth 240°, Tilt 2°
Tri-Sector with Variable Downtilt	`30; 150; 270`	`2.5; 4.0; 2.5`	✅ **Creates 3 CellSectors:** • Sector 1: Azimuth 30°, Tilt 2.5° • Sector 2: Azimuth 150°, Tilt 4.0° • Sector 3: Azimuth 270°, Tilt 2.5°
Dual Sector (2 Sectors)	`60; 240`	`3; 3`	✅ **Creates 2 CellSectors:** • Sector 1: Azimuth 60°, Tilt 3° • Sector 2: Azimuth 240°, Tilt 3°
Single Sector / Omni-directional	`0`	`0`	✅ **Creates 1 CellSector:** • Single antenna facing 0° with 0° tilt
❌ **Count Mismatch (Invalid)**	`0; 120; 240`	`2; 2`	❌ **SKIPPED:** 3 azimuths vs 2 beam tilts. Sector creation skipped.

Antenna Configuration	`Azimuth (°)`	`Beam Tilt (°)`	Sector Breakdown & Ingestion Behavior
❌ **Wrong Delimiter (Invalid)**	`0, 120, 240`	`2, 2, 2`	❌ **FAILED:** Uses commas `,` instead of semicolons `;`. Decimal parsing fails.

8. Blank Values vs. Dummy Placeholders

For optional parameters (e.g., `Zone`, `Barangay` if unincorporated, or inactive band columns):

- Leave the cell completely **BLANK (EMPTY)**.
- **DO NOT** type placeholder characters such as `"---"`, `"-"`, `"N/A"`, `"null"`, `"NONE"`, or `"#REF!"`.

9. Post-Upload Support

If your file encounters processing errors:

1. Review the progress logs in the **Cell** tab of the portal.
2. Note the batch number and reported record index.
3. Check whether the failure stems from missing coordinates, unrecognized provider names, or invalid non-numeric antenna heights.
4. Contact the system administrator with the file name and log excerpt for assistance.